

Ayman Mahfuz

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EDUCATION

The University of Texas at Austin

Aug 2023 - May 2027

Bachelor of Science, Computer Science, Concentration: AI and ML

Austin, TX

• GPA: 3.5/4.0

• Coursework: High Performance Computing, Natural Language Processing, Generative Visual Computing, Energy-Efficient Computing

EXPERIENCE

Arm

May 2025 - Aug 2026

Platform Validation Intern / Co-op – Software Engineering

- Reproduced silicon failures across three CPU-platform validation phases by building a production Python Bayesian optimizer for stress-ng configurations; owned development and deployment.
- Guided ML stress-test selection with CPU performance counters, using Python scripts to optimize cache-miss rate, memory bandwidth, and stall rate, which identified high-stress configurations for further validation
- Earned internal patent-filing approval and a 12-month co-op extension; presented production results to SVP and director leadership.

UT Austin – Texas Robotics & AI Lab

Jan 2025 - Jul 2026

Undergraduate Research Assistant – Robotics RL & LLM Evals

- Built a seeded, multi-turn evaluation harness to test whether a coin-game RL advisor transfers to Persuade for Good donation dialogues; served three Qwen variants with vLLM on TACC GPUs, recorded splits, seeds, model IDs, manifests, and 54 full JSONL trajectories, which provided clear evidence of transfer performance for each variant
- Accelerated policy-training iterations 2–3x by profiling GPU utilization and removing training-pipeline bottlenecks; scaled experiments to 5M+ episodes.
- Deployed 15 hierarchical reinforcement-learning policies to NAO robots and built C++/Python training environments within a 400K+ line codebase, contributing to the team's third-place finish at RoboCup 2025.

UT Austin – Center for Media Engagement

Aug 2023 - May 2025

Undergraduate Research Assistant – Software Engineering

- Built concurrent Selenium and REST ingestion workers with retry handling, enabling reliable processing of a 250M+ row BigQuery dataset containing 50M+ articles and 70M+ comments, which reduced ingestion failures and accelerated pipeline throughput
- Prevented duplicate experiment events and preserved round-robin assignments for 1,000+ participants through deterministic assignment logic, event deduplication, and structured logging.

UT Austin – Oden Institute

Aug 2023 - Jan 2025

Undergraduate Research Assistant – ML Systems

- Improved MRI-segmentation Dice score by 12 percentage points through convolutional and transformer model experiments with five-fold validation on 1,000+ scans using NVIDIA H100 GPUs.

PROJECTS

HelmPM - Multi-Tenant Project Management Platform

- Built a multi-tenant platform managing thousands of tickets shipped to multiple organizations, with a React frontend, FastAPI backend, PostgreSQL database, and automated Linear/Jira synchronization.
- Eliminated re-embedding of unchanged documents with Merkle-tree indexing, reducing API costs by 40%; implemented semantic, keyword, and exact-ID search across documents and code.

Activation Checkpoint Integrity in PyTorch - Independent Research

Python, PyTorch, CUDA

- Built a PyTorch audit that pairs original and recomputed tensors during activation checkpointing, localizes silent value divergence, and gates the optimizer before mismatched gradients can mutate training state, under Professor Mootaz Elnozahy.

Energy Profiling of LLM & Diffusion Inference

Python, PyTorch, vLLM, SLURM

- Ran 577 GH200 experiments across nine models and 55 configurations, measuring 24–32% lower LLM inference energy with FP8/INT8 and no measurable diffusion savings; built a vLLM/SLURM harness instrumented with NVML/Zeus and analyzed results using Welch's t-tests, Bonferroni correction, and roofline analysis.

Transformer Pretraining & Post-Training Pipeline (253M Parameters)

PyTorch, HPC

- Pretrained a 253M-parameter transformer on 600M tokens with RoPE, RMSNorm, and SwiGLU; built resumable mixed-precision HPC training with checkpoint recovery, supervised fine-tuning, and DPO.

SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, C

Backend/Web/Data: React, FastAPI, Flask, PostgreSQL, BigQuery, REST APIs, Pandas, NumPy, AWS

ML: PyTorch, Transformers, vLLM, LoRA/PEFT, PPO/MAPPO, RLHF/DPO, MCP, Harness Engineering

Systems/Cloud: CUDA, SLURM, NVML/Zeus, AWS, Docker, Linux, Distributed Systems